

PHYSICS

1) $\int_3^5 \frac{1}{2x+3} dx$ is equal to :-

(1) $\ln\left(\frac{13}{9}\right)$

(2) $\frac{1}{2}\ln\left(\frac{13}{9}\right)$

(3) $\frac{1}{2}\ln\left(\frac{15}{9}\right)$

(4) None

2) If $a \sin\theta + b \cos\theta = A \sin(\theta + \phi)$, then find A and ϕ .

(1) $\sqrt{a^2 + b^2}, \tan^{-1} \frac{b}{a}$

(2) $\sqrt{a^2 + b^2}, \tan^{-1} \frac{a}{b}$

(3) $(a + b), \tan^{-1} \frac{a}{b}$

(4) $\sqrt{a^2 + b^2}, -\tan^{-1} \left(\frac{a}{b}\right)$

3) If $y = x \log_e x$, then find $\frac{dy}{dx}$:-

(1) $1 - \log_e x$

(2) $1 + x$

(3) $\log_e x$

(4) $1 + \log_e x$

4) $\frac{d}{dx}(\log(\sin x^2))$ will be :-

(1) $\frac{\cos x^2}{\log(\sin x^2)}$

(2) $\frac{1}{x^2}(\sin x^2)$

(3) $2x \cot x^2$

(4) $2x \cot x$

5) If $y = \sin x \cos x$, then find out $\frac{dy}{dx}$:-

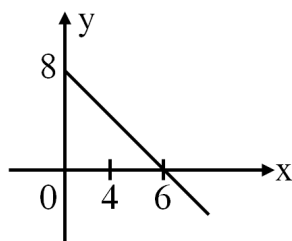
- (1) $\cos 2x$
- (2) $\frac{\cos 2x}{2}$
- (3) $\frac{\sin 2x}{2}$
- (4) $\sin^2 x - \cos^2 x$

6) If $y = 2x^2 - 12x$, then minimum value of y is:-

- (1) -18
- (2) 3
- (3) -3
- (4) 18

7)

The equation of straight line shown in figure is :



- (1) $6x + 8y = 15$
- (2) $4x + 3y = 18$
- (3) $2y + 6x = 7$
- (4) $3y + 4x = 24$

8)

Find $\log 32$ (given $\log 2 = 0.3$)

- (1) 0.5
- (2) 1.5
- (3) 2.3
- (4) 1.3

9)

Value of $\cos 106^\circ$ is :-

- (1) $\frac{7}{25}$
- (2) $\frac{-7}{25}$

(3) $\frac{24}{25}$

(4) $\frac{-24}{25}$

10) $\frac{d}{dx}(e^{10}) = \dots$

(1) $10e^9$

(2) e^{10}

(3) e^9

(4) 0

11)

If $\tan \theta = \frac{1}{\sqrt{5}}$ and θ lies in the first quadrant, the value of $\cos \theta$ is -

(1) $\sqrt{\frac{5}{6}}$

(2) $-\sqrt{\frac{5}{6}}$

(3) $\sqrt{\frac{1}{6}}$

(4) $-\sqrt{\frac{1}{6}}$

12)

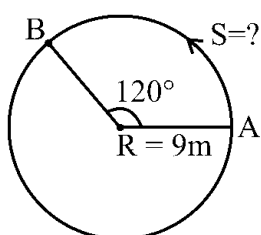
Value of $\cos 405^\circ$ is -

(1) $-\frac{1}{2}$

(2) $-\frac{1}{\sqrt{2}}$

(3) $\frac{1}{\sqrt{2}}$

(4) $\frac{\sqrt{3}}{2}$



13)

If radius $R = 9\text{m}$ and angle $\theta = 120^\circ$, then find arc length S .

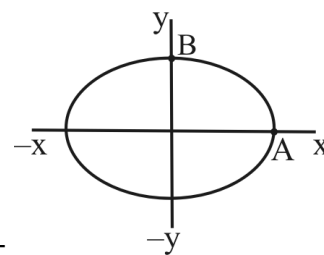
- (1) 1080 m
- (2) 13.33 m
- (3) 6π m
- (4) 9m

14) Find the sum of a series $1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \dots \infty$

- (1) $\frac{4}{3}$
- (2) $\frac{3}{2}$
- (3) $\frac{5}{2}$
- (4) $\frac{13}{9}$

15) The value of x on solving the quadratic equation $2x^2 - 7x + 5 = 0$ is :-

- (1) $\frac{5}{2}$ and 1
- (2) $\frac{3}{2}$ and 1
- (3) $-\frac{5}{2}$ and 1
- (4) $\frac{5}{2}$ and - 1



16) For ellipse $\frac{x^2}{9} + \frac{y^2}{25} = 1$, the co-ordinate of point A & B are respectively :-

- (1) (0, 3) & (5, 0)
- (2) (3, 0) & (0, 5)
- (3) (0, 9) & (25, 0)
- (4) (25, 0) & (0, 9)

17)

$\cos^2 15^\circ - \sin^2 15^\circ = \dots\dots\dots ?$

- (1) $\frac{1}{2}$
- (2) 1

(3) $\frac{\sqrt{3}}{2}$

(4) $\frac{\sqrt{3}-1}{2\sqrt{2}}$

18)

If $y = 2x^3 - 12x^2 + 36$, then for minimum value of y is :

(1) $x = 0$

(2) $x = 4$

(3) $x = 1$

(4) None

19) If $y = e^{\sin x}$, then $\frac{dy}{dx} = ?$

(1) $-\cos x \cdot e^{\sin x}$

(2) $e^{\cos x}$

(3) $e^{-\cos x}$

(4) None of these

20)

Which of following can be correct for any value of θ :-

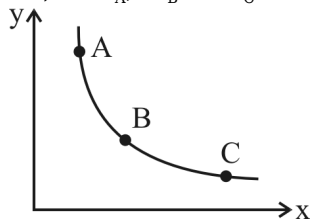
(1) $\sin \theta = -2$

(2) $\cos \theta = 2$

(3) $\tan \theta = 1000$

(4) All

21) If m_A , m_B & m_C are the slopes at points A, B and C respectively in the figure shown then :-



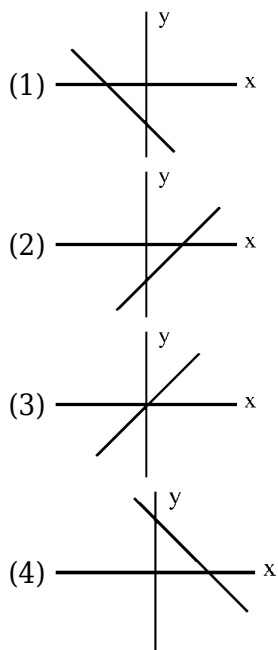
(1) $m_A > m_B > m_C$

(2) $m_A = m_B = m_C$

(3) $m_A = m_C < m_B$

(4) $m_A < m_B < m_C$

22) If $-3x - 5y = 15$ then $y - x$ graph is :-



23) If $y = ax^3 + bx + c \log_e x$ then find $\frac{dy}{dx} = ?$

- (1) $3ax^2 + bx + \frac{c}{x}$
- (2) $3ax^2 + b + c \log_e x$
- (3) $3ax^2 + 0 + \frac{c}{x}$
- (4) $3ax^2 + b + \frac{c}{x}$

24) If $y = \sin^2 x + \cos^2 x$, then $\frac{dy}{dx}$ is :-

- (1) $2\sin x + 2\cos x$
- (2) 1
- (3) 0
- (4) $2\sin x - 2\cos x$

25) If $y = 4x^2 + 5$ then slope of $y - x$ graph at $x = 2$ is :-

- (1) 21
- (2) 16
- (3) 8
- (4) 13

26) Which function remain unchanged after differentiation ?

- (1) e^{24}
- (2) e^x
- (3) Both (1) & (2)

(4) None of these

27) If $y = 2\sin^2 x$ then $\frac{dy}{dx}$ is :-

- (1) $\sin 2x$
- (2) $\cos 2x$
- (3) $2\sin 2x$
- (4) $2\cos 2x$

28) $\int_0^{\pi/4} \cos 2x = \dots$

- (1) 1
- (2) 3
- (3) $\frac{1}{2}$
- (4) $\frac{1}{4}$

29) $\int_0^2 3x^2 \cdot dx + \int_0^{\pi} \sin x \cdot dx = \dots?$

- (1) 8
- (2) 9
- (3) 10
- (4) $2 + \pi$

30) $(1+x)^{-n}$, where $x \ll 1$, can be written as :-

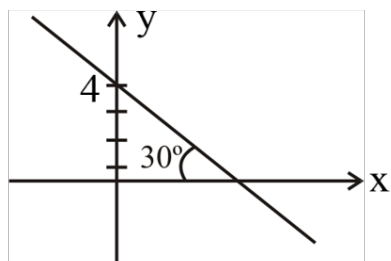
- (1) $1 + nx$
- (2) $1 - nx$
- (3) $x + n$
- (4) $x - n$

31) $\frac{1}{5} + \frac{1}{25} + \frac{1}{125} + \frac{1}{625} + \dots \infty = ?$

- (1) $\frac{1}{4}$
- (2) $\frac{5}{4}$
- (3) 1
- (4) Can't be determined

32) Value of $\sin 1^\circ$ is approximately equal to:-

- (1) 1
- (2) $\frac{\pi}{180}$
- (3) $\frac{180}{\pi}$
- (4) None of these



33) Equation of given line is :

- (1) $y = \sqrt{3}x + 4$
- (2) $y = -\sqrt{3}x + 4$
- (3) $y = \frac{1}{\sqrt{3}}x + 4$
- (4) $y = -\frac{1}{\sqrt{3}}x + 4$

34) Graph of $4x^2 + 4y^2 = 16$ can be given by is :-

- (1)
- (2)
- (3)
- (4)

35) If $y = x^{10}$ then find rate of change of y with respect to x at $x = 1$:-

- (1) $10x^9$
- (2) 10
- (3) x^9
- (4) 9

36)

Evaluate $\int \cos(5x - 3) dx =$

- (1) $-5 \sin(5x - 3) + c$
- (2) $5 \sin(5x - 3) + c$
- (3) $-\frac{1}{5} \sin(5x - 3) + c$
- (4) $\frac{1}{5} \sin(5x - 3) + c$

37) The value of $\int_0^3 (x^2 + 1) dx$ is :-

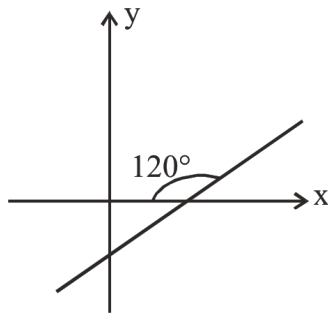
- (1) 10
- (2) 9
- (3) 12
- (4) 27

38) If radius of circle decreases as 0.2 cm/sec., then rate of change of its area (when radius of circle is 1cm) will be :-

- (1) $0.6\pi \text{ cm}^2/\text{sec}$
- (2) $0.4\pi \text{ cm}^2/\text{sec}$
- (3) $0.4 \text{ cm}^2/\text{sec}$
- (4) $0.2 \text{ cm}^2/\text{sec}$

39) Which of the following function remain unchanged after differentiation or integration?

- (1) $\sin x$
- (2) $\tan x$
- (3) $\sec x$
- (4) e^x



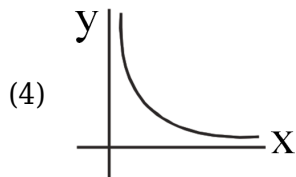
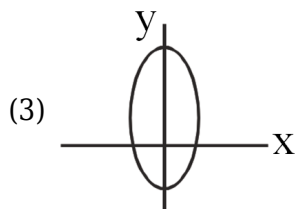
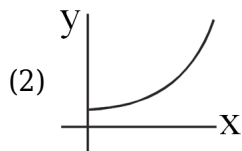
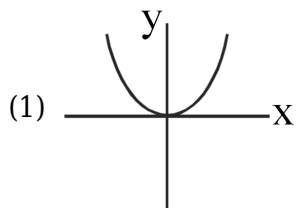
40) For given graph slope will be -

- (1) $\tan 30^\circ$
- (2) $\tan 120^\circ$
- (3) $\sqrt{3}$
- (4) Both (1) and (2)

41) $\frac{3\pi}{4}$ rad = ____?

- (1) 150°
- (2) 135°
- (3) 225°
- (4) 240°

42) The graph for $xy = 2$ will be :-



43) Find area bounded under the curve $y = 3x^2$ from $x = 0$ to $x = 3\text{m}$:-

- (1) 3 m^2

- (2) 9 m^2
 (3) 27 m^2
 (4) 81 m^2

44) $\int_4^2 dx$ is equal to :-

- (1) 2
 (2) -2
 (3) 6
 (4) 8

45)

If $\operatorname{cosec} \theta = \frac{5}{3}$ then $\tan \theta$ is :

- (1) $\frac{3}{4}$
 (2) $\frac{4}{3}$
 (3) $\frac{5}{4}$
 (4) $\frac{3}{5}$

CHEMISTRY

1) From the following statement choose the correct one :-

- (I) Kinetic energy decreases with increase in radius.
 (II) Total energy increases with increase in radius.

- (1) Statement-I is correct only.
 (2) Both statement-I and II are correct.
 (3) Both statement-I and II are incorrect.
 (4) Statement-II is correct only.

2) Calculate the energy (J/atom) associated with electronic transition from $n = 4$ to $n = 2$ in He^+ ion :-

- (1) $1.64 \times 10^{-18} \text{ J / atom}$
 (2) 13.6 J / atom
 (3) $2.18 \times 10^{-18} \text{ J / atom}$
 (4) $20.2 \times 10^{+20} \text{ J / atom}$

3) The radius of 2nd orbit of H atom will be :-

- (1) $0.529 \times \frac{2}{1} \text{ \AA}$
- (2) $0.529 \times \frac{4}{1} \text{ \AA}$
- (3) $0.529 \times \frac{1}{2} \text{ \AA}$
- (4) $0.529 \times \frac{1}{4} \text{ \AA}$

4) An increasing order for $\frac{e}{m}$ value for ${}^2_1\text{D}^+$, ${}^1_1\text{H}^+$, ${}^3_1\text{T}^+$ and ${}^4_2\text{He}^+$:-

- (1) $\text{He}^+ < \text{T}^+ < \text{D}^+ < \text{H}^+$
- (2) $\text{H}^+ < \text{T}^+ < \text{D}^+ < \text{He}^+$
- (3) $\text{T}^+ < \text{D}^+ < \text{H}^+ < \text{He}^+$
- (4) None of above

5) The ratio of kinetic energy to the total energy of electron :-

- (1) -1
- (2) +1
- (3) -2
- (4) +2

6) Which of the following are isodiaphers ?

- (1) ${}^{11}_5\text{B}$, ${}^{13}_6\text{C}$
- (2) ${}^{39}_{19}\text{K}$, ${}^{40}_{20}\text{Ca}$
- (3) ${}^3_1\text{H}$, ${}^4_2\text{He}$
- (4) ${}^{14}_7\text{N}$, ${}^{14}_6\text{C}$

7) Find the ratio of energies of 2nd and 3rd orbit of He^+ ion :-

- (1) $\frac{1}{4}$
- (2) $\frac{4}{1}$
- (3) $\frac{9}{4}$
- (4) $\frac{1}{2}$

8) Kinetic energy of e^- in second orbit of He^+ ion is 13.6 eV, then calculate kinetic energy of e^- in third orbit of He^+ ion :-

- (1) 6.04 eV
- (2) 3.02 eV
- (3) 13.6 eV
- (4) -13.6 eV

9) Calculate the wavelength of last line of lyman series of H-atom :-

- (1) 912 Å
- (2) 912×2 Å
- (3) $\frac{912}{2}$ Å
- (4) 912×4 Å

10) The angular momentum of electron in 3rd orbit of H-atom :-

- (1) $\frac{3h}{2\pi}$
- (2) $\frac{6h}{2\pi}$
- (3) $\frac{2h}{3\pi}$
- (4) $\frac{h}{2\pi}$

11) In a Bohr's model of H-atom, when an electron jumps from $n = 1$ to $n = 3$, how much energy will be emitted or absorbed per atom?

- (1) 1.93×10^{-18} J
- (2) 0.239×10^{-10} J
- (3) 2.15×10^{-11} J
- (4) 0.1936×10^{-10} J

12) Shortest wavelength in paschen series of Li^{2+} ion will be :-

- (1) $\frac{1}{R}$
- (2) $\frac{27R}{5}$
- (3) $30R$
- (4) $\frac{R}{8}$

13) In which of the following transition photon of maximum energy is absorbed :-

- (1) 3 to 2
- (2) 2 to 4
- (3) 1 to ∞

(4) 5 to 6

14) Calculate the 3rd orbit radius of H-atom :-

(1) 0.529 Å

(2) 4.761 Å

(3) 2.72 Å

(4) 5.429 Å

15) Calculate radius ratio of 2nd orbit to 3rd orbit of Li²⁺ ion :-

(1) 3 : 4

(2) 1 : 2

(3) 2 : 1

(4) 4 : 9

16) Calculate the energy of photon of wavelength 1000 Å :-

(1) 2×10^{-18} J

(2) 14×10^{-18} J

(3) $3 \times 10^{+18}$ J

(4) 2×10^{-6} J

17) If r_1 is the radius of the first orbit of hydrogen atom, then the radii of second, third and fourth orbits in term of r_1 are

(1) r_1^2, r_1^3, r_1^4

(2) $4r_1, 9r_1, 16r_1$

(3) $8r_1, 27r_1, 64r_1$

(4) $2r_1, 6r_1, 8r_1$

18) Calculate the energy in joule corresponding to light of wavelength 45 nm :
(Planck's constant $h = 6.63 \times 10^{-34}$ Js; speed of light $c = 3 \times 10^8$ ms⁻¹)

(1) 6.67×10^{15}

(2) 6.67×10^{11}

(3) 4.42×10^{-15}

(4) 4.42×10^{-18}

19) Which of the following statement is **not** correct about the characteristics of cathode rays?

(1) They start from the cathode and move towards the anode.

(2) They travel in straight line in the absence of an external electrical or magnetic field.

(3) Characteristics of cathode rays do not depend upon the material of electrodes in cathode ray tube.

(4) Characteristics of cathode rays depend upon the nature of gas present in the cathode ray tube.

20) The nucleus of the atom consists of:

- (1) Proton and neutron
- (2) Proton and electron
- (3) Proton and electron
- (4) Proton, neutron and electron

21) Electromagnetic radiation with the maximum wavelength is

- (1) Ultraviolet
- (2) Microwave
- (3) X-ray
- (4) Infrared

22) Carbon-12 atom has

- (1) 6 electron, 6 proton, 6 neutron
- (2) 12 electron, 6 proton, 6 neutron
- (3) 12 electron, 6 proton, 7 neutron
- (4) 6 electron, 12 proton, 18 neutron

23) The energies E_1 and E_2 of two radiations are 25 eV and 50 eV respectively. The relation between their wavelengths i.e. λ_1 and λ_2 will be

- (1) $\lambda_1 = \frac{1}{2} \lambda_2$
- (2) $\lambda_1 = \lambda_2$
- (3) $\lambda_1 = 2\lambda_2$
- (4) $\lambda_1 = 4\lambda_2$

24) The ratio of the energy of a photon of wavelength 3000Å to that of a photon of wavelength 6000Å is:

- (1) $\frac{1}{2}$
- (2) 2
- (3) 3
- (4) $\frac{1}{3}$

25) Select the **correct** Statements from the following :

- A. Atoms of all elements are composed of two fundamental particles.
- B. The mass of the electron is 9.10939×10^{-31} kg.
- C. All the isotopes of a given elements show same chemical properties.
- D. Protons and electrons are collectively known as nucleons.
- E. Dalton's atomic theory, regarded the atom as an ultimate particle of matter.

Choose the **correct** answer from the options given below.

- (1) C,D and E only
 (2) A and E only
 (3) B,C and E only
 (4) A,B and C only

26) If number of protons in $X_2 = 16$ number of electron in X^{2-} will be:-

- (1) 16
 (2) 8
 (3) 10
 (4) 6

27)

The **impossible** value of angular momentum of electron in H-atoms.

- (1) $3.5 \frac{h}{2\pi}$
 (2) $2.5 \frac{h}{\pi}$
 (3) $1.5 \frac{h}{\pi}$
 (4) $0.5 \frac{h}{\pi}$

28) Ionisation energy of He^+ is

- (1) 13.6 eV
 (2) 3.4 eV
 (3) 54.4 eV
 (4) -13.6 eV

29) The radius of 3rd Bohr's orbit of H-atom (Given - radius of first bohr orbit = r_0)

- (1) $9r_0$
 (2) $3r_0$
 (3) $15r_0$
 (4) $18r_0$

30) If an electron jump from 2nd orbit to 3rd orbit in Li^{2+} . Then calculate change in energy of electron will be.

- (1) 5.4 eV
 (2) 17 eV
 (3) 1.8 eV
 (4) 10 eV

31) If the table of atomic masses were established with Carbon (C-12) atom and assigned value 150 unit, then the mass of magnesium (Mg-24) atom would be, approximately.

- (1) 12 unit
- (2) 300 unit
- (3) 25 unit
- (4) 24 unit

32) If mass of neutron is doubled, mass of proton is halved and mass of electron is doubled then find out new atomic weight of ${}^6_{14}\text{C}$.

- (1) 12 unit
- (2) 14 unit
- (3) 6 unit
- (4) 19 unit

33) If the KE of e^- in second and third orbit of Li^{2+} ion is x and y respectively then

- (1) $x = y$
- (2) $x > y$
- (3) $x < y$
- (4) None of above

34) Yellow, light emitted from a lamp has a wavelength (λ) of 580 nm. Calculate the wave number of yellow light.

- (1) $1.927 \times 10^5 \text{ cm}^{-1}$
- (2) $1.724 \times 10^4 \text{ cm}^{-1}$
- (3) $1.0 \times 10^{-5} \text{ cm}^{-1}$
- (4) $2.147 \times 10^{-4} \text{ cm}^{-1}$

35) The frequency of photon is directly proportional to its

- (1) Wavelength
- (2) Energy
- (3) Valency
- (4) None of above

36) Time period of revolution by e^- in H atom is directly proportional to

- (1) $\frac{n^3}{Z^2}$
- (2) $\frac{Z^3}{n^2}$
- (3) $\frac{Z}{n}$

(4) $\frac{n^2}{z}$

37) According to Bohr's theory, which of the following correctly represents the variation of energy and radius of an electron in n^{th} orbit of H-atom.

(1) $E \propto \frac{1}{n^2}, r \propto \frac{1}{n^2}$

(2) $E \propto \frac{1}{n^2}, r \propto n^2$

(3) $E \propto n^2, r \propto n^2$

(4) $E \propto n, r \propto \frac{1}{n}$

38) Which transition has minimum wavelength (λ)

(1) $2 \rightarrow 1$

(2) $3 \rightarrow 1$

(3) $4 \rightarrow 1$

(4) $4 \rightarrow 3$

39) Which of the following series of transitions in the spectrum of hydrogen atom fall in visible region.

(1) Lyman series

(2) Balmer series

(3) Paschen series

(4) Brackett series

40) What is the 3rd excitation energy for He^+ .

(1) 13.6 eV

(2) 1.51 eV

(3) 0.85 eV

(4) 51 eV

41) Angular momentum of 2nd Bohr orbit of H-atom is

(1) $\frac{h}{2\pi}$

(2) $\frac{h}{\pi}$

(3) $\frac{3h}{2\pi}$

(4) $\frac{2h}{\pi}$

42) The ratio of charge to mass would be greater for

- (1) Proton
- (2) Electron
- (3) Neutron
- (4) Alpha particle

43) If the energy difference between the ground state of an atom and excited state is 4.4×10^{-19} J. The wavelength of photon required to produce this transition.

- (1) 4.5×10^{-7} m
- (2) 4.5×10^{-7} nm
- (3) 4.5×10^{-7} Å
- (4) 4.5×10^{-7} cm

44) Which of the following electromagnetic radiation has least value of frequency

- (1) Micro waves
- (2) Radio waves
- (3) X-rays
- (4) I.R. rays

45) The ratio of radius of 2nd orbit of He⁺ ion to 3rd orbit of H atom will be -

- (1) $\frac{4}{25}$
- (2) $\frac{2}{9}$
- (3) $\frac{3}{10}$
- (4) $\frac{3}{20}$

BIOLOGY

1)

Heparin work as a :

- (1) Vasoconstrictor
- (2) Allergy
- (3) Anti - coagulant
- (4) Irritation

2) Strongest Cartilage is :-

- (1) White fibrous cartilage
- (2) Elastic cartilage

- (3) Hyaline cartilage
- (4) Calcified cartilage

3) In tissue, the group of cell should always be similar in :

- (1) Structure
- (2) Function
- (3) Origin
- (4) All of above

4) Condition necessary for the formation of organ system :

- (1) Presence of all four basic type of tissue
- (2) Chemical intraction
- (3) Physical intraction
- (4) All of these

5) Which is the matrix producing cell in connective tissue ?

- (1) Fibroblast
- (2) Lymphocytes
- (3) Mast cell
- (4) Both 1 and 3

6) Major inorganic part of the bone is :

- (1) $\text{Ca}_3(\text{PO}_4)_2$
- (2) CaCO_3
- (3) $\text{Mg}_3(\text{PO}_4)_2$
- (4) All are equal in quantity

7) Structurally how many parts are present in the mature long bone ?

- (1) 1
- (2) 2
- (3) 3
- (4) 4

8) Squamous epithelium occurs in inner lining of

- (1) Convulated tubules of nephrons
- (2) Alveoli of lungs
- (3) Pancreatic duct
- (4) Ciliary body

9) Ligament connects

- (1) Bone to bone
- (2) Bone to muscle
- (3) Skin to muscle
- (4) Fibrous tissue

10) Choose the correctly matched pair :-

- (1) Areolar tissue - loose connective tissue
- (2) Cartilage - loose connective tissue
- (3) Tendon - specialised connective tissue
- (4) Adipose tissue - Dense connective tissue

11) Cells of squamous epithelium are :-

- (1) Long/Tall, column like
- (2) Cube like
- (3) Thin, flat, tiles like
- (4) Tall with elongated nuclei

12) Epithelium which lines the urinary bladder is :-

- (1) Stratified cuboidal epithelium
- (2) Stratified columnar epithelium
- (3) Pseudostratified epithelium
- (4) Transitional epithelium

13) Type of tissue that is involved in formation of glands:-

- (1) Epithelium
- (2) Muscular
- (3) Nervous
- (4) All of the above

14) Choose the correctly matched pair regarding epithelial in the given below :

- (1) Tubular part nephron - Cuboidal epithelium
- (2) Inner lining of bronchioles - Squamous epithelium
- (3) Inner lining of salivary duct - Ciliated epithelium
- (4) Moist surface of buccal cavity - Glandular epithelium

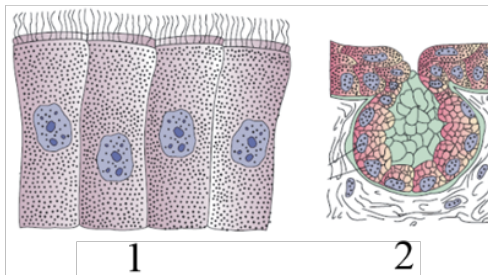
15) Match the column and select the correct answer-

	Column - I		Column - II
A.	Hyaline cartilage	(i)	Intervertebral disc

B.	White fibrous cartilage	(ii)	Tracheal rings
C.	Elastic cartilage	(iii)	Tip of nose
		(iv)	Nasal septum
		(v)	Ear pinna

- (1) A - i, iv; B - iii; C - i, v
 (2) A - i, ii; B - iv; C - iii, v
 (3) A - ii, iv; B - i; C - iii, v
 (4) A - ii, v; B - i, iv; C - iii

16) A student was given a sample of two tissues. He observes the tissues under the microscope and



draws their figures (1 and 2) as shown below. :- tissues (1 and 2).

- (1) 1 : Columnar cells bearing cilia;
 2 : Unicellular glandular epithelium
 (2) 1: Cuboidal cells bearing cilia;
 2: Multicellular glandular epithelium
 (3) 1: Compound cells bearing cilia;
 2: Unicellular glandular epithelium
 (4) 1: Columnar cells bearing cilia;
 2: Multicellular glandular epithelium

17) A student was given a sample of tissue. He observe and concludes the following characters

- (a) The cells are composed of a single layer of flat cells
 (b) The cells have flat centrally located nucleus
 (c) It is found in the lining of blood vessels and alveoli

Based on the above features identify the epithelium

- (1) cuboidal epithelium
 (2) columnar epithelium
 (3) squamous epithelium
 (4) Glandular epithelium

18) Which of the following statement (s) regarding cell junctions is / are correct?

- (1) **Tight junctions** help in stoppage of substances from leaking across a tissue
 (2) **Adhering junctions** perform cementing to keep neighbouring cells together
 (3) **Gap junctions** facilitate the cells to communicate with each other by connecting the cytoplasm of adjoining cells, for rapid transfer of ions, small molecules and sometimes big molecules

(4) All of the above

19) Read the following statements carefully

(a) They have a hard and nonpliable ground substance

(b) They support and protect softer tissues and organs

(c) Osteocytes are present in spaces called lacunae.

Which of the following type of tissue is being described by the above statements?

(1) Blood

(2) Cartilage

(3) Bone

(4) Neurons

20) Match the type of connective tissue given in column A with their examples given column B and choose the correct option

	Column A (type of connective tissue)		Column B (examples)
a	Loose connective tissue	i	Tendons and ligaments
b	Dense regular tissue	ii	Skin
c	Dense irregular tissue	iii	Cartilage , bones and blood
d	Specialized connective tissue	iv	Fibroblasts, macrophages and mast cells

	a	b	c	d
1	i	iv	ii	iii
2	i	iv	iii	ii
3	iv	i	ii	iii
4	iv	ii	i	iii

(1) 1

(2) 2

(3) 3

(4) 4

21) Choose the correct option w.r.t. the location and functions of brush border columnar epithelium.

	Location	Function
(1)	PCT	Filtration
(2)	Small intestine	Filtration

(3)	Large intestine	Secretion
(4)	Small intestine	Absorption

(1) 1

(2) 2

(3) 3

(4) 4

22) Find out the wrongly matched pair

(1) Squamous epithelium - Urinary bladder

(2) Columnar epithelium - Stomach

(3) Cuboidal epithelium - Germinal epithelium

(4) Ciliated epithelium - Fallopian tube

23) Epithelium in type II pneumocyte

(1) Ciliated cuboidal

(2) Cuboidal epithelium

(3) Ciliated columnar

(4) Columnar

24) Match the following columns.

	Column I		Column II
A.	Areolar tissue	1.	Fat cells
B.	Adipose tissue	2.	Osteocytes
C.	Ligament	3.	Loose connective tissue
D.	Bone	4.	Dense regular connective tissue

Codes:

	A	B	C	D
(1)	3	1	4	2
(2)	1	2	3	4
(3)	4	3	2	1
(4)	2	1	4	3

(1) 1

(2) 2

(3) 3

(4) 4

25) **Assertion (A) :-** Epithelial tissue are avascular.

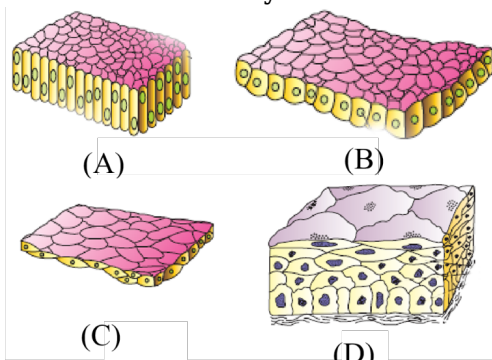
Reason (R) :- Cells are compactly packed in epithelial tissue.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (3) (A) is true but (R) is false.
- (4) Both (A) and (R) are false

26) Outer most covering of cartilage is called _____ which is composed of _____ fibrous connective tissue :-

- (1) perichondrium, yellow
- (2) periostium, white
- (3) perichondrium, white
- (4) periostium, yellow

27) The four sketches (A, B, C and D) given below represent four different types of animal tissues. Which one of these is correctly identified in the option given along with its correct location and

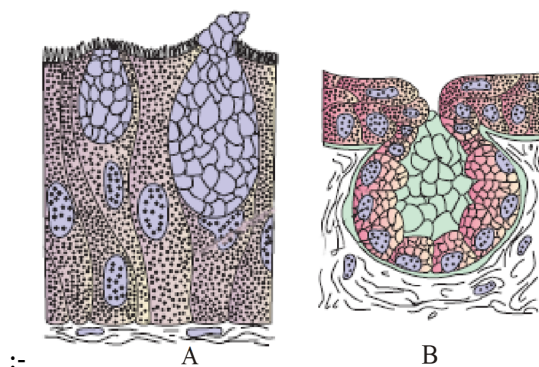


function?

		Tissue	Location	Function
1.	B	Simple squamous epithelium	Fallopian tube	Transport of gamete
2.	C	Simple cuboidal epithelium	Wall of blood vessels and air sac of lungs	Diffusion boundary
3.	D	Compound epithelium	Skin	Protection
4.	A	Simple columnar epithelium	Tubular part of nephron	Secretion

- (1) 1
- (2) 2
- (3) 3
- (4) 4

28) Identify the glands (A) and (B) shown below and select the right option for location and function



		Gland	Location	Function
1.	A	Unicellular Gland	Alveoli	Secrete Saliva
2.	B	Multicellular gland	Oesophagus	Secrete enzyme
3.	A	Multicellular gland	Alimentary Canal	Secrete Mucous
4.	B	Multicellular gland	Buccal Cavity	Secrete Saliva

- (1) 1
(2) 2
(3) 3
(4) 4

29)

Read the following four statements (A - D) :-

- (A) Power of regeneration is highest in epithelium
(B) Basement membrane is secreted by both epithelium and muscular tissue
(C) During embryonic development, epithelium originated first
(D) Epithelium has abundant vascularisation.

How many above statements are correct ?

- (1) four
(2) one
(3) two
(4) three

30) How many of following is composed of columnar epithelium and it's modification ?

Liver, Gall bladder, PCT, collecting duct, Ependymal epithelium, mesothelium, Stomach, intestine

- (1) 3
(2) 5
(3) 4
(4) 2

31) Given below are two statements -

Statement I : Bones have a solid and pliable ground substances rich in calcium salts and collagen fibre.

Statement II : Most of cartilages in vertebrate embryos are replaced by bone in adults.

In the light of the above statements, choose the correct answer from the option given below.

- (1) Both statement-I and statement-II are incorrect.
- (2) Statement-I is correct but statement-II is incorrect.
- (3) Statement-I is incorrect but statement - II is correct.
- (4) Both statement-I and statement-II are correct.

32) Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Some of the columnar or cuboidal cells get specialised for secretion.

Reason (R) : Columnar epithelium is commonly found in duct of glands and tubular part of nephron in kidneys.

In the light of the above statements, choose the correct answer from the option given below.

- (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
- (2) (A) is correct but (R) is not correct.
- (3) (A) is not correct but (R) is correct.
- (4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

33) Consider the following statements (A) to (C) and select the correct option stating which ones are TRUE (T) and which ones are FALSE(F).

- A. Stratified squamous epithelium covers moist surfaces like buccal cavity.
- B. Fibroblast store fat in adipose tissue.
- C. Urinary bladder is lined by a stratified epithelium.

- (1) A-F; B-T; C-T
- (2) A-T; B-F; C-F
- (3) A-F; B-T; C-F
- (4) A-T; B-F; C-T

34) **Statement-I :** Goblet cells is the example of unicellular glandular epithelium.

Statement-II : Exocrine gland secrete mucus, saliva, earwax, oil, hormones, milk, digestive enzymes.

- (1) Both statements are correct
- (2) Statement I is correct Statement II is incorrect
- (3) Statement I is incorrect Statement II is correct
- (4) Both statements are incorrect

35) How many of the following statements are incorrect for Bone :-

- (A) Bones have hard and non-pliable ground substance
- (B) Matrix is rich in collagen fibres
- (C) Osteoblasts are present in lacunae
- (D) Bone is a specialised connective tissue
- (E) They serve weight-bearing function

- (1) 4
- (2) 3

(3) 1

(4) 2

36) Consider the following statements.

- (a) Organ system Contributes to the survival of the body as a whole.
 - (b) All complex animals consist of only one type of tissue.
 - (c) The epithelial tissue does not face a body fluid.
 - (d) Squamous epithelium is involved in forming a diffusion boundary.
- Select the option with incorrect statements only.

(1) (a),(b) and (c)

(2) (b),(c) and (d)

(3) (a),(c) and (d)

(4) (b) and (c)

37) Maximum part of voice box is composed of :-

- (1) Elastic cartilage
- (2) White fibrous cartilage
- (3) Calcified bone
- (4) Hyaline cartilage

38) Haversian system is also called as :

- (1) Osteon
- (2) Volkmann canal
- (3) Osteoblast
- (4) Mast cell

39) Which of the following is correct match of cartilage with its feature ?

- (1) Hyaline cartilage - rich in collagen fibres
- (2) White fibrous cartilage - Pubic-symphysis
- (3) Elastin cartilage - strongest cartilage
- (4) Calcified cartilage - highly flexible cartilage

40)

Tendon is made up of:-

- (1) Mainly collagen fibres
- (2) Elastic connective tissue fibres
- (3) Fibrous cartilage
- (4) In elastic and elastic connective tissue

41) Which of the following is a transparent tissue?

- (1) Fibrous cartilage
- (2) Hyaline cartilage
- (3) Tendon
- (4) Ligament

42) Which fibre form dense network?

- (1) Collagen
- (2) Elastin
- (3) Reticular
- (4) All of these

43) True about transitional epithelium is :-

- (1) Non-stretchable
- (2) Basement membrane is thick and highly complex
- (3) Present in distal part of male urethra
- (4) Inner most layer of cell is composed of cube like cells

44) Ground substance of which specialised connective tissue rich in calcium salts and collagen fibres ?

- (1) Blood
- (2) Cartilage
- (3) Bone
- (4) Lymph

45) Fibre which is highly resistant to chemicals is:-

- (1) Collagen fibre
- (2) Elastin fibre
- (3) Reticulin fibre
- (4) Collagen and reticulin fibre

46) What type of epithelium forms epidermis ?

- (1) simple squamous epithelium
- (2) pseudostratified epithelium
- (3) compound epithelium
- (4) stratified cuboidal epithelium

47)

The cells of germinal epithelium are

- (1) Cuboidal

- (2) Ciliated
- (3) Columnar
- (4) Squamous

48)

Mast cells secrete

- (1) Heparin
- (2) Histamine
- (3) Serotonin
- (4) All of these

49)

Histiocyte is a connective tissue cell which takes part in

- (1) Secretion
- (2) Phagocytosis
- (3) Fibre production
- (4) Matrix production

50)

Which of the following is a loose connective tissue ?

- (1) Blood
- (2) Tendon
- (3) Areolar tissue
- (4) Nervous tissue

51)

Collagen fibres are secreted by

- (1) Fibroblasts
- (2) Mast cells
- (3) Astrocytes
- (4) Macrophages

52) Stratified squamous epithelium

- (1) Outer most layer squamous and inner most is cuboidal
- (2) Outer most layer cuboidal and inner most is squamous
- (3) Outer most layer columnar and inner most is cuboidal
- (4) Outer most layer cuboidal and inner most is columnar

53) Which of the following is known as spongy connective tissue?

- (1) Dense fibrous connective tissue
- (2) Adipose connective tissue
- (3) Areolar connective tissue
- (4) Reticular fibrous connective tissue

54) Scavenger cell of connective tissue proper is called as

- (1) Fibroblast cell
- (2) Mast cell
- (3) Mesenchymal cell
- (4) Macrophages

55) In which of the following locations are ciliated columnar epithelial cells are found?

- (1) Brush bordered membrane in tubular part of nephron.
- (2) Brush border membrane in intestine
- (3) Fallopian tube and ventricles of brain.
- (4) Air sacs of lungs

56) The most widely distributed tissue in the body of complex animals :-

- (1) Epithelial tissue
- (2) Connective tissue
- (3) Muscular tissue
- (4) Nervous tissue

57) Which of the following cell junction is found in epithelial tissue and help to stop substances from leaking across tissue?

- (1) Tight junction
- (2) Gap junction
- (3) Adhering junction
- (4) Interdigititation

58) All the following structures are lined by transitional epithelium except :-

- (1) Urinary bladder
- (2) Renal pelvis
- (3) Ureter
- (4) Oesophagus

59) Goblet cells are the examples of :-

- (1) Unicellular glands

- (2) Multicellular glands
- (3) Endocrine glands
- (4) Both (2) and (3)

60) Which epithelium provide protection against chemical and mechanical stresses and also play a limited role in secretion and absorption

- (1) Simple columnar epithelium
- (2) Transitional epithelium
- (3) Compound epithelium
- (4) Glandular epithelium

61) Basement membrane is secreted by :-

- (1) Epithelium tissue only
- (2) Connective tissue only
- (3) Both epithelium and connective tissue
- (4) Epithelium and muscular tissue both

62) The epithelium of proximal convoluted tubule (PCT) of nephron in the kidney has :-

- (1) Cilia
- (2) Steriocilia
- (3) Microvilli
- (4) Kinocilia

63) Nature of salivary gland is :-

- (1) Unicellular, endocrine
- (2) Multicellular, endocrine
- (3) Multicellular, exocrine
- (4) Unicellular, exocrine

64) The endothelium of blood vessels is made up of simple _____?

- (1) Cuboidal epithelium
- (2) Squamous epithelium
- (3) Columnar epithelium
- (4) Ciliated columnar epithelium

65) The tissue which act as basal structure for epithelium :-

- (1) Areolar tissue
- (2) Muscular tissue
- (3) Bones
- (4) Cartilage

66) What type of epithelium is more than one layer thick?

- (1) Endothelium
- (2) Stratified
- (3) pseudostratified
- (4) Tessellated

67) Which of the following type of cell junction is not found in animal tissue :-

- (1) Desmosome
- (2) Tight junction
- (3) Gap junction
- (4) Plasmodesmata

68) The secretion of saliva, mucus, ear wax, oil and milk is associated with :-

- (1) Exocrine gland
- (2) Endocrine gland
- (3) Duct less glands
- (4) Adipose tissue

69) Which connective tissue is present beneath the skin and often it serves as a support frame work for epithelium ?

- (1) Dense regular connective tissue
- (2) Dense irregular connective tissue
- (3) Areolar connective tissue
- (4) Mucoid connective tissue

70) Epithelium which forms a diffusion boundary is:-

- (1) Cuboidal epithelium
- (2) Columnar epithelium
- (3) Squamous epithelium
- (4) Compound epithelium

71) Which one of the following is **not** a connective tissue ?

- (1) Bone
- (2) Cartilage
- (3) Blood
- (4) Muscles

72) Select the odd one w.r.t. secretions of glands which is released through ducts.

- (1) Saliva

- (2) Oil
- (3) Hormone
- (4) Milk

73) Following is responsible for elongation of bone :

- (1) Shaft
- (2) Diaphysis
- (3) Epiphysis
- (4) Metaphysis

74) Find out the epithelial tissue in which nuclei are located at the base :-

- (1) Columnar epithelium tissue
- (2) Glandular epithelium tissue
- (3) Compound epithelium tissue
- (4) Squamous epithelium tissue

75) Which type of fibres are stained by silver-salts ?

- (1) White collagen fibres
- (2) Yellow elastic fibres
- (3) Arzyrophills fibres
- (4) All of these

76)

Protein present in cartilage is :-

- (1) Cartilagin
- (2) Ossein
- (3) Chondrin
- (4) Oesein

77) Pseudostratified epithelium is modification of.

- (1) Squamous
- (2) Cuboidal
- (3) Columnar
- (4) Stratified columnar

78) Free surface of epithelial tissue faces:-

- (1) Basement membrane
- (2) Body fluid
- (3) Outside environment

(4) Both (2) and (3)

79) Animal cell differ from plant cell on the basis of :-

- (1) Mitochondria
- (2) Cell membrane
- (3) Plastid
- (4) Golgibody

80) Select the wrong pair :-

Types of cells		Shape of cells
(1)	Red blood cells	Round and biconcave
(2)	Columnar epithelial cells	Long and narrow
(3)	Nerve cell	Unbranched and long
(4)	Mesophyll cells	Round and oval

- (1) 1
- (2) 2
- (3) 3
- (4) 4

81) Which one of the following is not a component of cell wall of algae :

- (1) Cellulose
- (2) Galactans
- (3) Hemicellulose
- (4) Mannans

82) A structure that connect the cytoplasm of neighbouring cells, and another which holds or glues the different neighbouring cell together. These are

- (1) Cell wall and middle lamella, respectively
- (2) Plasmodesmata and middle lamella, respectively
- (3) Middle lamella and desmosomes, respectively
- (4) Middle lamella and plasmodesmata, respectively

83) How many of the following organelles are membrane bound organelles ?
Lysosomes, Ribosomes, Mitochondria, Vacuole

- (1) 2
- (2) 3
- (3) 4
- (4) 1

84) Which of the following is true for primary cell-wall :-

- (1) It is capable of growth and gradually diminishes as cell matures
- (2) It is not capable of growth and gradually diminishes as the cell matures
- (3) It is capable of growth and remains intact as cell-matures
- (4) It is capable of growth and denature rapidly.

85) The middle lamella consists of :

- (1) Suberin
- (2) Cellulose
- (3) Lignin
- (4) Calcium pectate

86) Cell theory which was given by Schleiden and Schwann, did not explain as to how the new cells are formed ? modified the hypothesis of cell theory to give it a final shape ?

- (1) Nageli
- (2) R. Virchow
- (3) Robert Hooke
- (4) C.P. Swanson

87) **Assertion:** Cell is the fundamental structural and functional unit of all living organism

Reason: Anything less than a complete structure of a cell don't ensure independent living.

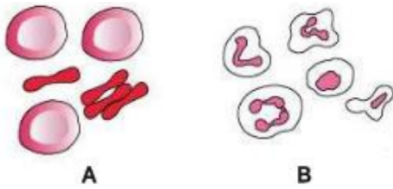
- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

88) Match column-I (scientists) with column-II (discovery) and select the correct option.

	Column-I		Column-II
(A)	Leeuwenhoek	(I)	First saw and described a living cell
(B)	Robert Brown	(II)	Presence of cell wall is unique to plant cells
(C)	Schleiden	(III)	Discovered the nucleus
(D)	Schwann	(IV)	All plants are composed of different kind of cells

- (1) A - I; B - III; C - IV; D - II
- (2) A - I; B - III; C - II; D - IV
- (3) A - III; B - I; C - IV; D - II
- (4) A - I; B - IV; C - II; D - III

89) According to given figure which of the following option is correct ?



- (1) A-Red blood cells - Round and biconvex
- (2) B - White blood cells - Amoeboid
- (3) A - Red blood cells - about $3\mu\text{m}$ in diameter
- (4) All of these

90) Which of the following statement is incorrect ?

- (1) Unicellular organisms are capable of independent existence
- (2) Any structure less than a complete structure of a cell does not ensure independent living
- (3) Matthias schleiden, a German zoologist, Proposed that body of animals are composed of cells and product of cells
- (4) Schleiden and Schwann together formulated the cell theory

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	1	4	3	1	1	4	2	2	4	1	3	3	2	1	2	3	2	4	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	4	1	4	3	2	2	3	3	3	2	1	2	4	3	2	4	3	2	4	3
Q.	41	42	43	44	45															
A.	2	4	3	2	1															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	2	1	2	1	1	1	3	1	1	1	1	1	3	2	4	1	2	4	4	1
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	2	1	3	2	3	3	1	3	1	2	2	4	2	2	2	1	2	3	2	4
Q.	86	87	88	89	90															
A.	2	2	1	2	2															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	3	1	3	1	4	1	2	2	1	1	3	4	1	1	3	4	3	4	3	3
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	4	1	2	1	1	3	3	4	3	2	3	2	4	2	3	4	4	1	2	1
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	3	4	3	2	3	1	4	2	3	1	1	3	4	3	2	1	4	1	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	3	3	2	1	2	4	1	3	3	4	3	4	1	3	3	3	4	3	3
Q.	171	172	173	174	175	176	177	178	179	180										
A.	3	2	2	1	4	2	1	1	2	3										

PHYSICS

$$1) \int_3^5 \frac{1}{2x+3} = \frac{1}{2} [\ln(2x+3)]_3^5$$

$$= \frac{1}{2} \ln\left(\frac{13}{9}\right)$$

2)

Let $a = A \cos \phi$, $b = A \sin \phi$

Then,

$$a \sin \theta + b \cos \theta = A \sin \theta \cos \phi + A \cos \theta \sin \phi$$

$$= A \sin(\theta + \phi)$$

where, $A = \sqrt{a^2 + b^2}$ and

$$\tan \phi = \frac{b}{a}$$

$$\Rightarrow \phi = \tan^{-1}\left(\frac{b}{a}\right)$$

$$3) y = x \log_e x$$

$$\frac{dy}{dx} = x \left(\frac{1}{x}\right) + \log_e x (1)$$

$$\frac{dy}{dx} = 1 + \log_e x$$

$$5) y = \frac{\sin 2x}{2} \Rightarrow \frac{dy}{dx} = \frac{2 \cos 2x}{2} = \cos 2x$$

6)

$$y = 2x^2 - 12x$$

$$\frac{dy}{dx} = 4x - 12 = 0 \Rightarrow x = +3$$

$$\frac{d^2y}{dx^2} = 4 > 0$$

so, minimum value of $y = 2(3)^2 - 12 \times 3 = -18$

7)

$$y = mx + c \text{ where, } m = \frac{0-8}{6-0} = -\frac{4}{3} \text{ and}$$

$$c = 8 \text{ (at } x = 0, y = c = 8)$$

Therefore, equation of straight line,

$$y = -\frac{4}{3}x + 8$$

$$\Rightarrow 3y + 4x = 24$$

8)

$$\log 32 = \log 2^5 = 5 \log 2 = 5 \times 0.3 = 1.5$$

9)

$$\cos 2A = 2 \cos^2 A - 1$$

$$13) S = 9 \times \frac{2\pi}{3} = 6\pi$$

$$14) S_{\infty} = \frac{a}{1-r}$$

$$S_{\infty} = \frac{a}{1-1/3} = \frac{3}{2}$$

$$15) 2x^2 - 7x + 5 = 0 \quad 2x^2 - 2x - 5x + 5 = 0$$

$$2x(x-1) - 5(x-1) = 0$$

$$(x-1)(2x-5) = 0$$

$$x = 1 \text{ and } x = \frac{5}{2}$$

16)

Question Explanation:

The problem gives an ellipse equation $\frac{x^2}{9} + \frac{y^2}{25} = 1$ and asks for the coordinates of two points (A and B) where the ellipse intersects the axes.

Concept :

Coordinate geometry

Formula Used:

The general form of the ellipse equation is:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

Where a is the semi-major axis and b is the semi-minor axis. Coordinates of points are found by substituting a: x = 0 or y = 0 for axis intersections.

Calculation :

For the given equation, $a^2 = 9$ (so a = 3) and $b^2 = 25$ (so b = 5).

- For Point A (on the x-axis), set $y = 0$:

$$\frac{x^2}{9} = 1 \Rightarrow x = \pm 3$$

Thus, A = (3, 0).

- For Point B (on the y-axis), set $x = 0$:

$$\frac{y^2}{25} = 1 \Rightarrow y = \pm 5$$

Thus, B = (0, 5).

Therefore, the correct answer is Option 2: (3, 0) & (0, 5).

$$17) \cos^2 15^\circ - \sin^2 15^\circ$$

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

$$\square \cos^2 15^\circ - \sin^2 15^\circ$$

$$= \cos 30^\circ$$

$$= \sqrt{3}/2$$

$$18)$$

$$y = 2x^3 - 12x^2 + 36$$

$$\frac{dy}{dx} = 6x^2 - 24x + 0 = 6x^2 - 24x$$

For maxima or minima :-

$$\frac{dy}{dx} = 0$$

$$6x^2 - 24x = 0$$

$$6x(x - 4) = 0$$

$$x = 0 \text{ and } x = 4$$

$$\frac{d^2y}{dx^2} = 12x - 24$$

$$\text{at } x = 0 \Rightarrow \frac{d^2y}{dx^2} = -24 < 0 \rightarrow \text{maxima}$$

$$\text{at } x = 4 \Rightarrow \frac{d^2y}{dx^2} = 24 > 0 \rightarrow \text{minima}$$

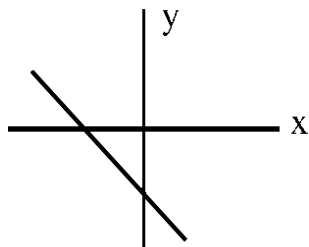
$$19) \frac{d}{dx} (e^{\sin x}) = (\cos x)e^{\sin x}$$

$$20) -1 \leq \sin \theta \leq +1$$

$$+1 \leq \cos \theta \leq +1$$

$$-\infty < \tan \theta < +\infty$$

choose options according to Range.



22)

$$-3x - 5y = 15$$

$$5y = -3x - 15$$

$$y = -\frac{3}{5}x - 3$$

Here $\therefore m = -\frac{3}{5} \quad c = -3$

23) **1. Explanation (30 words):**

We are asked to find the derivative of the function $y = ax^3 + bx + c \log_e(x)$ which involves differentiating each term separately.

2. Concept (20 words):

To differentiate a function, apply basic differentiation rules such as the power rule and the derivative of $\log_e(x)$.

3. Formula:

A. 1. The derivative of x^n is nx^{n-1} .

B. The derivative of $\log_e(x)$ is $\frac{1}{x}$.

4. Calculation (Minimum Steps):

Differentiate each term:

1. The derivative of ax^3 is

$$\frac{d}{dx}(ax^3) = 3ax^2$$

2. The derivative of bx is:

$$\frac{d}{dx}(bx) = b$$

3. The derivative of $c \log_e(x)$ is:

$$\frac{d}{dx}(c \log_e(x)) = \frac{c}{x}$$

Thus, the derivative of $y = ax^3 + bx + c \log_e(x)$ is:

$$\frac{dy}{dx} = 3ax^2 + b + \frac{c}{x}$$

24) **Explain** - find $\frac{d}{d\pi} (\sin^2 x + \cos^2 x)$

Concept - Differentiation

Formula

$$\sin^2 \pi + \cos^2 \pi = 1$$

$$\frac{d[\text{constaw}]}{dx} = 0$$

Calculation

$$\frac{d}{dx} \sin^2 x + \cos^2 x$$

$$\frac{d(1)}{dx}$$

25) **1. Asking About:**

The question is asking for the slope of the graph of the equation $y = x$ at a specific point, when $x = 2$.

2. Concept:

The slope of a graph at any point is the rate of change of the function at that point. This is calculated by taking the derivative of the function with respect to x (denoted as $\frac{dy}{dx}$), which gives us the slope of the tangent to the graph at any point.

3. Formula:

$$\text{Slope} = \frac{dy}{dx}$$

4. Solution / Explanation:

The given equation is $y = 4x^2 + 5$.

To find the slope of the graph at any point, first differentiate y with respect to x :

$$\frac{dy}{dx} = \frac{d}{dx}(4x^2 + 5) = 8x$$

Put $x = 2$ gives slope = 16

5. Final Answer:

The slope of the graph at $x = 2$ is **16**. Therefore, the correct option is **Option 2**.

27)

By chain rule

$$\frac{dy}{dx} = 4 \sin x \cos x = 2 \sin 2x$$

28) **Explanation :**

The definite integral of $\cos(2x)$ from 0 to $\pi/4$

Concept : Integration

Formula :

$$\text{Integration of } \cos(ax) : \int \cos(ax) dx = \frac{1}{a} \sin(ax) + C$$

$$\int_a^b f(x) dx = F(b) - F(a)$$

Definite Integral : $\int_a^b$

where $F(x)$ is the antiderivative of $f(x)$

Calculation : Evaluating from 0 to $\pi/4$

$$\begin{aligned} \left[\frac{1}{2} \sin(2x) \right]_0^{\pi/4} &= \frac{1}{2} \sin\left(2 \times \frac{\pi}{4}\right) - \frac{1}{2} \sin(2 \times 0) \\ &= \frac{1}{2} \sin\left(\frac{\pi}{2}\right) - \frac{1}{2} \sin(0) = \frac{1}{2} \times 1 - \frac{1}{2} \times 0 = \frac{1}{2} \end{aligned}$$

Final Answer: The definite integral is $1/2$

Correct Option: Option 3 ($1/2$)

$$\begin{aligned}
 & 29) \int_0^2 3x^2 dx + \int_0^{\pi} \sin x dx \\
 &= 3 \left(\frac{x^3}{3} \right)_0^2 + (-\cos x)_0^{\pi} \\
 &= 8 + (\cos 0 - \cos \pi) \\
 &= 8 + (1 - (-1)) \\
 &= 8 + 2 = 10
 \end{aligned}$$

30) **1. Explanation (30 words):** We are asked to find the binomial expansion approximation for $(1+x)^{-n}$ when $x \gg 1$. The binomial expansion helps approximate powers of binomials.

2. Concept (20 words):

For large values of, the binomial expansion of $(1+x)^{-n}$ simplifies to an approximation involving powers of x .

3. Formula:

The binomial expansion for $(1+x)^{-n}$:

$$(1+x)^{-n} \approx 1 - nx \quad (\text{for } x \gg 1)$$

4. Calculation (Minimum Steps):

For $(1+x)^{-n}$ where $x \gg 1$, we use the binomial approximation:

$$(1+x)^{-n} \approx 1 - nx \quad (\text{for large } x)$$

32)

$$\begin{aligned}
 \sin 1^\circ &= \sin \left(1^\circ \times \frac{\pi}{180^\circ} \right) \\
 &\text{for angle to be small} \\
 \sin \theta &\approx \theta
 \end{aligned}$$

$$\begin{aligned}
 33) \text{ Slope} &= -\tan \theta = \frac{y}{x} \\
 \Rightarrow -\frac{1}{\sqrt{3}} &= \frac{y}{x} \Rightarrow y = -\frac{1}{\sqrt{3}}x \\
 \text{intercept} &= +4 \\
 \text{So } y &= -\frac{1}{\sqrt{3}}x + 4
 \end{aligned}$$

34)

Concept Based:

This equation represents a circle. By simplifying the equation, we can check if it matches the standard forms of a circle's equation $x^2 + y^2 = r^2$.

Formula Used:

The standard equation for a circle $x^2 + y^2 = r^2$.

Calculation:

Given $4x^2 + 4y^2 = 16$, divide both sides by 4:

$$x^2 + y^2 = 4$$

This is the standard form of the equation of a circle with radius $r = 2$, centered at the origin.

Thus, the graph of the given equation is a circle with radius 2, centered at the origin.

35) $\frac{dy}{dx} = 10x^9$

at $x = 1$

$$\frac{dy}{dx} = 10$$

36) **1. Question Explanation :**

We are asked to evaluate the integral of $\cos(5x - 3)dx$ and find the correct result among the given options.

2. Concept used :

trigonometric function with a linear expression inside the cosine function.

3. Formula Used :

The integral of $\cos(ax + b) dx$ is given by :

$$\int \cos(ax + b) dx = \frac{1}{a} \sin(ax + b) + C$$

where a is the coefficient of x inside the cosine, and b is a constant.

4. Calculation :

Comparing with the standard form :

Here $a = 5$, $b = -3$

Applying the formula:

$$I = \frac{1}{5} \sin(5x - 3) + C$$

Final Answer :

Option (4) : $\frac{1}{5} \sin(5x - 3) + C$

38)

Given $\frac{dr}{dt} = 0.2 \text{ cm/s}$

Area of circle $A = \pi r^2$

$$\begin{aligned} \Rightarrow \frac{dA}{dt} &= \pi \cdot \frac{dr^2}{dt} \\ &= \pi \cdot 2r \cdot \frac{dr}{dt} \end{aligned}$$

when $r = 1 \text{ cm}$

$$\frac{dA}{dt} = \pi \times 2 \times 1 \times 0.2 = 0.4\pi \text{ cm}^2/\text{s}$$

39)

e^x remain same after differentiation & integration.

40)

θ is measured from +ve x-axis $\Rightarrow \tan 60^\circ = m = \sqrt{3}$

42) **1. Explanation (30 words):**

We are asked to describe the graph of the equation $xy = 2$. This equation represents a hyperbola. To understand it better, we analyze its structure and behavior.

2. Concept (20 words):

The equation $xy = 2$ represents a rectangular hyperbola, which has two branches and asymptotes along the coordinate axes.

3. Formula:

The equation $xy = 2$ can be rewritten as:

$$y = \frac{2}{x}$$

This is the standard form of a rectangular hyperbola.

$$44) y = \int_4^2 dx = [x]_4^2 = 2 - 4 = -2$$

CHEMISTRY

46) $\Rightarrow$ KE decreases if n increases

$\Rightarrow$ Total E. increases if n increases

$$47) E = 2.18 \times 10^{-18} \times 2^2 \left[\frac{1}{2^2} - \frac{1}{4^2} \right] \text{ J / atom}$$

$$E = 2.18 \times 10^{-18} \times \frac{3}{4} \text{ J / atom}$$

$$E = 1.64 \times 10^{-18} \text{ J / atom}$$

$$48) r = 0.529 \times \frac{n^2}{2} \text{ \AA}$$

$$49) D^+ = \frac{1}{2}; \quad T^+ = \frac{1}{3}$$

$$H^+ = \frac{1}{1}; \quad He^+ = \frac{1}{4}$$

50)

$$KE = -TE$$

51) Isodiaphers have same $(n-p)$ value.

$$52) \frac{E_2}{E_3} = \frac{-13.6 \times \frac{2^2}{2^2} \text{ eV}}{-13.6 \times \frac{2^2}{3^2} \text{ eV}} = \frac{9}{4}$$

$$53) \text{ KE} = 13.6 \times \frac{2^2}{3^2} \text{ eV} = 6.04 \text{ eV}$$

$$54) \frac{1}{\lambda} = R(1)^2 \left[\frac{1}{1^2} - \frac{1}{\infty^2} \right]$$

$$\frac{1}{\lambda} = R; \lambda = 912 \text{ \AA}$$

$$55) mvr = \frac{nh}{2\pi}$$

56)

Question Explanation:

The question asks us to calculate the energy absorbed when an electron in a Bohr atom transitions from $n = 1$ to $n = 3$.

Given Data:

Initial state: $n_1 = 1$

A. Final state: $n_2 = 3$

Concept

Energy of an Electron in a Bohr Orbit: The energy of an electron in the n th orbit of a hydrogen-

like atom is given by: $E_n = -\frac{2\pi^2 m Z^2 e^4}{n^2 h^2}$ where:

A. h = planck's constant

B. Energy Difference: The energy absorbed or emitted during a transition is the difference

$$\Delta E = E_2 - E_1 = 2\pi^2 m Z^2 e^4 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) / h^2$$

between the final and initial energy levels:

C. Rydberg Constant: We can also express this in terms of the Rydberg constant (R):

$$\Delta E = R h c \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

where $R = 2.18 \times 10^{-18} \text{ J}$ or 2.18×10^{-11}

ergs(for hydrogen) h = Planck's constant c = speed of light.

A. We need to convert Joules to ergs if necessary. $1 \text{ Joule} = 10^7 \text{ ergs}$

D: Mathematical Calculation

1. Calculate the energy difference:

$$\Delta E = Rhc \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$\Delta E = (2.18 \times 10^{-11} \text{ ergs}) \left(\frac{1}{1^2} - \frac{1}{3^2} \right)$$

$$\Delta E = (2.18 \times 10^{-11} \text{ ergs}) \left(1 - \frac{1}{9} \right)$$

$$\Delta E = (2.18 \times 10^{-11} \text{ ergs}) \left(\frac{8}{9} \right)$$

$$\Delta E = 1.9377 \times 10^{-11} \text{ ergs}$$

E: Final Answer

The energy absorbed when an electron jumps from $n = 1$ to $n = 3$ is approximately **0.193×10^{-10} ergs..**

The option is **(4) 0.193×10^{-10} ergs.**

F: Question Level Moderate

57) Shortest wavelength line ($\infty \rightarrow 3$)

$$\frac{1}{\lambda} = Rz^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$\frac{1}{\lambda} = R(9) \left(\frac{1}{9} - \frac{1}{\infty^2} \right)$$

$$\lambda = \frac{1}{R}$$

58) Max energy absorbed in 1 to ∞

$$59) r = 0.529 \times \frac{3^2}{1} \text{ \AA}$$

$$r = 4.761 \text{ \AA}$$

$$60) \frac{r_2}{r_3} = \frac{0.529 \times \frac{2^2}{3} \text{ \AA}}{0.529 \times \frac{3^2}{3} \text{ \AA}} = \frac{4}{9}$$

$$61) E = \frac{hc}{\lambda} = \frac{2 \times 10^{-25}}{10^{-7}} = 2 \times 10^{-18} \text{ J}$$

62)

Explanation

The radius of an orbit in a hydrogen atom is proportional to the square of the principal

quantum number (n).

Concept:

Radius (r_n) = $r_1 \times n^2$, where r_1 is the radius of the first orbit and n is the orbit number.

Calculation:

A. Radius of 2nd orbit (r_2) = $r_1 \times (2^2) = 4r_1$

B. Radius of 3rd orbit (r_3) = $r_1 \times (3^2) = 9r_1$

C. Radius of 4th orbit (r_4) = $r_1 \times (4^2) = 16r_1$

Answer Option 2, ($4r_1$, $9r_1$, $16r_1$)

63)

Explanation

A. Calculate the energy in joule corresponding to light of wavelength.

Given Data

A. Wavelength (λ) = 45 nm = 45×10^{-9} m

B. Planck's constant (h) = 6.63×10^{-34} Js

C. Speed of light (c) = 3×10^8 ms⁻¹

Concept

A. Energy of a photon (E) = hc/λ

Calculation

A. $E = (6.63 \times 10^{-34} \text{ Js} \times 3 \times 10^8 \text{ ms}^{-1}) / (45 \times 10^{-9} \text{ m})$

B. $E = (19.89 \times 10^{-26} \text{ Jm}) / (45 \times 10^{-9} \text{ m})$

C. $E = 4.42 \times 10^{-18} \text{ J}$

Answer option 4, (4.42×10^{-18} J).

64)

Explanation

- A. Cathode rays are streams of electrons emitted from the cathode in a vacuum tube. Their properties were crucial in discovering the electron.

Concept

- A. Cathode rays do originate from the cathode and move towards the anode.
- B. They do travel in straight lines unless influenced by external fields.
- C. They are fundamentally electrons, so their nature doesn't change with electrode material.
- D. However, the residual gas in the tube can affect their behavior (e.g., through collisions).

Answer option 4, (Characteristics of cathode rays depend upon the nature of gas present in the cathode ray tube).

65)

A: Question Explanation

The question asks us to identify the components of the atomic nucleus. We need to recall the basic structure of an atom.

B: Given Data

- A. The question asks what the nucleus of an atom consists of.

C: Concept

- A. Atomic Nucleus: The nucleus is the dense, positively charged center of an atom.
- B. Protons: Positively charged particles located in the nucleus.
- C. Neutrons: Neutral (no charge) particles located in the nucleus.

D. Electrons: Negatively charged particles that orbit the nucleus.

D: Final Answer

The nucleus of an atom consists of protons and neutrons.

The correct option is (1).

E: Question Level

Easy

66)

Explanation

A. Electromagnetic Spectrum:

- A. The electromagnetic spectrum arranges electromagnetic radiation by wavelength and frequency.
- B. Wavelength and frequency are inversely related: longer wavelength means lower frequency, and vice versa.

B. Wavelength Ordering:

- A. From longest to shortest wavelength: Radio waves, Microwaves, Infrared, Visible light, Ultraviolet, X-rays, Gamma rays.

C. Therefore:

- A. Microwaves have a longer wavelength than Ultraviolet, X-rays, and Infrared.

Answer option 2, (Microwave).

67)

Explanation

- A. The atomic number of an element indicates the number of protons.
- B. In a neutral atom, the number of electrons equals the number of protons.

Concept

- A. The mass number (for carbon-12, it's 12) is the sum of protons and neutrons.
- A. Carbon has an atomic number of 6, so it has 6 protons(6 electrons).
- B. Carbon-12 has a mass number of 12. To find the number of neutrons, subtract the number of protons from the mass number: $12 - 6 = 6$ neutrons.

Answer option 1, (6 electron, 6 proton, 6 neutron).

68)

Explanation

- A. Relation between wavelengths by using energies of different photons.

Given Data

A. $E_1 = 25 \text{ eV}$

B. $E_2 = 50 \text{ eV}$

Concept

- A. Energy of a photon (E) is inversely proportional to its wavelength (λ): $E = hc/\lambda$, where h is Planck's constant and c is the speed of light.

Calculation

- A. Relationship:

A. $E_1 = hc/\lambda_1$

B. $E_2 = hc/\lambda_2$

- B. Ratio:

A. $E_2/E_1 = (hc/\lambda_2) / (hc/\lambda_1) = \lambda_1/\lambda_2$

B. $50 \text{ eV} / 25 \text{ eV} = \lambda_1 / \lambda_2$

C. $2 = \lambda_1 / \lambda_2$

C. Wavelength relation:

A. $\lambda_1 = 2\lambda_2$

Answer option 3, ($\lambda_1 = 2\lambda_2$).

69) By Planck's theory

$$E \propto \frac{1}{\lambda}$$

$$\frac{E_1}{E_2} = \frac{\lambda_2}{\lambda_1} = \frac{6000}{3000} = \frac{2}{1}$$

70) It is statement based question.

Statements B, C & E are correct.

(B) Mass of the electron is $9.10939 \times 10^{-31} \text{ kg}$

(C) All the isotopes of given elements show same chemical properties.

(E) Dalton's atomic theory, regarded the atom as an ultimate particle of matter.

71) No. of protons in $x = 8$

No. of e^- in $x^{2-} = 8 + 2 = 10 e^-$

72) Angular momentum

$$mvr = \frac{nh}{2\pi} \text{ [n is a positive integer]}$$

73) Ionisation energy = $13.6 Z^2 \text{ eV}$.

74) radius of bohr orbit = $0.529 \frac{n^2}{Z} \text{ \AA}$

$$\begin{aligned} 75) \quad \Delta E &= 13.6 Z^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right] \text{ eV} \\ &= 13.6 \times 3^2 \left[\frac{1}{2^2} - \frac{1}{3^2} \right] \text{ eV} \\ &= 13.6 \times 3^2 [0.139] \text{ eV} \\ &= 17.01 \text{ eV} \end{aligned}$$

76) Ratio of old masses = Ratio of New masses.

$$\frac{12}{24} = \frac{150}{M_g}$$

$$M_g = 300 \text{ unit}$$

77) New mass of neutron + New mass of proton

$$8 \times 2 + 6 \times \frac{1}{2} = 19 \text{ unit}$$

78) KE of e^- is $13.6 \times \frac{Z^2}{n^2} \text{ eV}.$

79) $\lambda = \frac{1}{\bar{\nu}}$

$$580 \text{ nm} = \frac{1}{\bar{\nu}}$$

$$\bar{\nu} = \frac{1}{580 \times 10^{-7} \text{ cm}} = 1.724 \times 10^4 \text{ cm}^{-1}$$

83) lyman series lines will have maximum energy $\Rightarrow$ minimum wavelength.

85) 3rd SE = $n = 4$ to $n = \infty$

$$\Delta E = 13.6 \times 2^2 \left[\frac{1}{4^2} - \frac{1}{\infty^2} \right]$$

$$= \frac{13.6 \times 4}{16} \text{ eV} = 3.4 \text{ eV}$$

86) $mvr = \frac{nh}{2\pi}$

88) $\Delta E = \frac{he}{\lambda}$

$$4.4 \times 10^{-19} \text{ J} = \frac{2 \times 10^{-25} \text{ J.m}}{\lambda}$$

$$\lambda = 4.5 \times 10^{-7} \text{ m}$$

90) $\frac{r_{2\text{He}^+}}{r_{3\text{H}}} = \frac{0.529 \times \frac{2^2}{2} \text{ \AA}}{0.529 \times \frac{3^2}{1} \text{ \AA}} = \frac{4}{18} = \frac{2}{9}$

BIOLOGY

93) Module#2 Pg.#75

94) Module#2 Pg.#75, NCERT Pg.#100

95) Modul#2 Pg.#89

96) Module#2 Pg.#98

104) NCERT XIth Pg. # 101

117)

Old NCERT XI Page # 101

118) **Old NCERT - XI - Page no. 102**

121) **Old NCERT - XI - Page no. 104**

122) **Old NCERT - XI - Page no. 101, 102**

124) NCERT XI, Pg. # 102

134) NCERT Pg.# 104

141)

The correct answer is **Option 1: Fibroblasts.**

Fibroblasts are the primary cell type responsible for synthesizing and secreting collagen fibers. Collagen is the most abundant protein in the human body and is a major component of connective tissues like tendons, ligaments, and skin.

142)

The correct answer is **1. Outer most layer squamous and inner most is cuboidal.**

149)

NCERT Pg 102

159) **Solution/Explanation/Calculation:**

Areolar tissue is a loose connective tissue found beneath the skin and acts as a support framework for epithelium.

Correct Answer is : Areolar connective tissue.

Hence, option (3) is correct

169)

NCERT-XI Pg#88

170) NCERT Pg.# 89

171)

NCERT Pg.# 94

Explanation:

The cell wall of algae is primarily composed of cellulose, galactans (such as agar and carrageenan), and mannans. Hemicellulose is not typically a major component of algal cell walls; it is more commonly found in the cell walls of higher plants.

172)

NCERT XI Pg # 94

173) NCERT XI (Eng.), Pg.# 88

174)

NCERT Pg.# 94

Asking about: The characteristics of the primary cell wall in plant cells.

Concept: Understanding the structure and function of the primary cell wall and its role in plant growth.

Solution:

- A. Primary cell wall: This is the first cell wall layer formed during cell growth. It is relatively thin and flexible, allowing the cell to expand.
- B. Growth: The primary cell wall is capable of growth, which is essential for cell elongation and expansion during development.
- C. Maturation: As the cell matures, a secondary cell wall may be deposited inside the primary cell wall. The primary cell wall itself does diminish as the cell matures.

It is capable of growth and gradually diminishes as the cell matures: This is completely correct.

Therefore, the correct statement is that the primary cell wall is capable of growth and

gradually diminishes as cell matures

Answer: 1

175) NCERT Pg. # 94

176) NCERT XI Pg. # 88

177) NCERT XII Page No. # 87

178)

NCERT Pg.# 88

Answer: 1.

Solution:

Let's match the scientists in Column-I with their discoveries in Column-II:

(A) Leeuwenhoek: He was the first to observe and describe living cells (like bacteria and protozoa). So, A matches with (I).

(B) Robert Brown: He discovered the nucleus of the cell. So, B matches with (III).

(C) Schleiden: He concluded that all plants are composed of cells. So, C matches with (IV).

(D) Schwann: He concluded that animal cells have a cell membrane and that plant cells have a cell wall. He also stated that the presence of a cell wall is unique to plant cells. So, D matches with (II).

Therefore, the correct matching is:

A - I

B - III

C - IV

D - II

The answer is 1

179) NCERT Page no. 89

180) NCERT-XI Pg. # 88